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Graded Assignment • 30 min

Due Mar 18, 11:59 PM CET

1. Which of the following is true about the basic processes of evolution: 1 point

- Natural selection is a completely random process.
- Individual variation within a species cannot be inherited from generation to generation.
- Characteristics seen in individual members of a population rarely show variation that would lead to differential survival.
- Individual *fitness*, or capacity for survival and reproduction within a population, may depend on the specific environment the population inhabits.

2. DNA mutations (changes in base sequences on the DNA molecule): 1 point

- are the source for genetic variation within a population and are the raw material for natural selection
- can create inheritable variation in a population only when these mutations occur in somatic cells of the individual
- can not be used to estimate relationships between different species
- do not contribute to the ability of a population to adapt to changes in the environment

3. All of the following are true of genetic drift EXCEPT: 1 point

- Genetic drift can increase the frequency of an allele within a population due to chance variation in the percentage of individuals that survive and reproduce.
- Genetic drift may occur when a catastrophic event results in a small surviving population with alleles that may not reflect the occurrence of alleles in the population as a whole prior to the event.
- Genetic drift explains the way a new mutation might increase in frequency due to differences in survival or reproductive success in the environment.
- Genetic drift can include mutations that affect phenotypic characters that have no effect on survivability of a population if the character no longer has a function, such as the eyes of some animals that live in caves.

4. Male ornamentation, such as large antlers in male stags that may be particularly attractive to females or advantageous in fighting off other males, is an example of: 1 point

- natural selection that results in the evolution of features that are advantageous to the individual but not necessarily an adaptation to the external environment
- sexual selection for traits that help females attract mates
- a trait that will result in the improved survival of the individual male stag
- genetic drift that changes the frequencies of genes to promote large antlers

5. When examining variation in phenotype within a population, all of the following are true EXCEPT: 1 point

- variance in a population implies that values for a specific characteristic sort around a mean number
- when there is a perfect genetic relationship (heritability) between midparent and offspring mean values for a particular trait, then the slope of the correlation line will be equal to "one"
- when there is no genetic relationship (heritability) between midparent and offspring mean values for a particular trait, then the slope of the correlation line will be equal to "one"
- heritability of traits seen in the wild populations are measurable quantities and can be determined mathematically

6. Which of the following best explains phenoptypic expression as seen in an individual? 1 point

- It is a reflection of both the genetic inheritance and environmental influences to which it is exposed.
- It is a reflection of neither genetic inheritance nor environmental conditions.
- It is a reflection of the genetic inheritance of the individual.
- It is a reflection of the environmental conditions to which the individual is exposed.

7. In the study of Galapagos finches by Drs. Peter and Rosemary Grant: 1 point

- Bill depth was found to have very low (close to zero) heritability, showing no correlation of bill depth to the genetics of the original island finch population.
- Bill depth was found to have high heritability (90%), which explains the rapid rate of natural selection of bill depth as the finches populated different islands.
- The high heritability of bill depth can explain data showing that although the environment is different from island to island in the Galapagos, the bill characteristics of finches remained the same throughout the islands.
- The low heritability of bill depth accounts for the maintenance and stability of distinct bill depth variations from island to island.

8. Peppered moth variation in Europe: 1 point

- was the result of a mutation in the melanin pigment that occurred in the 1960's and gave rise to a light colored allele which avoided predation by birds as it rested on light colored trunks
- in the 1930's favored the selection of the light morphotype of the moth because it was the form that was visible against the darkened tree trunks and this improved sexual selection for light pigmented moths
- favored a decrease in melanism (black moths) as the coal dust was cleaned out of the air and tree trunks became cleaner allowing for predation of darker moths in the population
- was due to the persistence of two morphotypes of the moth with equal distribution of moth coloration throughout the 20th century

9. Research into a hantavirus outbreak in New Mexico demonstrated that: 1 point

- The hantavirus was very genetically pliable and demonstrated a new trans-species virus capable of using humans as a vector.
- The maintenance of type specimen collections such as those housed at the University of New Mexico are not assets for use when encountering outbreaks of unknown virulent disease agents.
- The outbreak was related to the movement of mice carrying a new genotype of hantavirus particle into houses and infecting humans who came in contact with the mice.
- Hantavirus phylogeny and rodent phylogeny show a co-evolutionary relationship that originated in Asia and has continued to evolve in American environments.

10. An adaptation is an outcome of the selective process: 1 point

- that contributes to the overall fitness of individuals allowing them to leave more offspring in future generations
- that can be consciously manipulated by the organism
- that does not contribute to the overall fitness of individuals
- the function of which cannot be tested experimentally